

# Amir Mirzai Golpayegani

📍 Calgary, AB, Canada

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## Summary

M.Sc. Computer Science graduate with experience building Python/C++ systems for scientific data processing, data pipelines, machine learning, and backend APIs. Skilled in Python, SQL/PostgreSQL, Flask/Django-style web services, Linux, AWS-based data workflows, PyTorch, debugging, documentation, and turning complex research datasets into reliable software.

## Education

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| <b>M.Sc. in Computer Science</b><br><i>University of Calgary</i>               | GPA: <b>3.94/4.0</b> | Sep 2022 – Feb 2026<br><i>Calgary, Canada</i> |
| <b>B.Sc. in Computer Engineering</b><br><i>Sharif University of Technology</i> | GPA: <b>3.69/4.0</b> | Sep 2016 – Feb 2021<br><i>Tehran, Iran</i>    |

## Technical Skills

**Programming:** Python, C++, JavaScript, SQL

**ML / AI:** PyTorch, Scikit-learn, TensorFlow, CNNs, LLM workflows, AI agents, model evaluation

**Data / Geospatial:** NumPy, Pandas, SciPy, SQL, Google Earth Engine API, Discrete Global Grid System (DGGS), GDAL, Rasterio, SNAP

**Backend / Engineering:** Django, REST APIs, PostgreSQL, Node.js, Docker, Git, Linux, AWS, GitHub Actions

**Visualization / Automation:** Streamlit, Polyscope, OpenGL, GLSL, n8n, OpenClaw, SearXNG, Telegram API

## Work Experience

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| <b>Research Software Developer</b><br><i>BigGeo / Mitacs, University of Calgary</i> | Sep 2022 – Feb 2026<br><i>Calgary, Canada</i> |
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- Built C++/Python software for scalable processing, validation, storage, retrieval, and analysis of scientific Earth observation data.
- Automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and satellite data APIs.
- Developed maintainable DGGS-based data pipelines to convert satellite imagery into structured tensor files.
- Improved encoding performance with C++ multithreading, reducing 10-tensor processing from 233s to 26s.
- Reduced storage by 38% for scaled Canada-wide data with 1,112 GeoTIFF tiles and 16K+ DGGS tensors.

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| <b>Back-End Developer</b><br><i>Asr Gooyesh Pardaz</i> | Jun 2020 – Jul 2022<br><i>Tehran, Iran</i> |
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- Developed Django REST APIs for NLP, text-to-speech, and speech-to-text products backed by PostgreSQL.
- Built authentication features including registration, Google login, password reset, and email/phone verification.
- Implemented backend logic for payment-service integration and storing plan/payment records in relational database tables.
- Contributed to a WebRTC-based chatbot system using GPT-style responses and FAQ data to answer client questions.

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| <b>Deep Learning Teaching Assistant</b><br><i>Neuromatch Academy</i> | Jul 2026<br><i>Remote, United States</i> |
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- Mentored an international pod of ~15 students through Neuromatch Academy's intensive three-week Deep Learning course.
- Facilitated hands-on PyTorch tutorials spanning MLPs, CNNs, attention/transformers, diffusion generative models, and reinforcement learning.
- Guided project groups through end-to-end deep learning research projects, from question formulation and literature review to model implementation and final presentations.

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| <b>Teaching Assistant</b><br><i>Department of Computer Science, University of Calgary</i> | Dec 2022 – Apr 2025<br><i>Calgary, Canada</i> |
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- **Developing Big Data Applications:** Mentored +25 students on AWS-based data pipelines, cloud storage, debugging, and scalable data workflows.
- **Working with Data at Scale:** Supported +20 students with large-scale data processing, data pipelines, and high-volume data analysis.
- **Database Management Systems:** Assisted +75 students with SQL, database design, course projects, grading, and exam support.

## Research and Technical Projects

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| <b>Spatiotemporal Earth Observation Data System Using DGGS</b><br><i>First-author publication   M.Sc. thesis   C++, Python, DGGS, Wavelets, B-splines, Time-Series, Data Retrieval</i> | Sep 2022 – Feb 2026 |
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- Designed a DGGs-based framework for real-time multiresolution management of satellite imagery.
- Developed a tensor-based storage model for direct access to arbitrary regions at multiple resolutions.
- Implemented a triangular wavelet scheme to support multiresolution retrieval without image-pyramid redundancy.
- Built B-spline temporal approximation to compress time-series data and retrieve missing timestamps locally.
- Retrieved 105,489 DGGs cells per time-series query in 13.9s, compared with about 220s using global retrieval.

**DEM Super-Resolution Framework**

May 2024 – Feb 2026

Python, PyTorch, CNNs, Scientific Imaging, Google Earth Engine API, Polyscope, Data Processing and Visualization

- Trained deep ResNet models for 4× DEM super-resolution, reconstructing 30m elevation data from 120m inputs.
- Built a Streamlit pipeline using Rasterio to parallelize GEE DEM tile downloads and systematically collect and preprocess mountainous terrain data worldwide.
- Implemented PyTorch training, validation, and evaluation workflows with terrain-aware metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors.
- Developed a Polyscope-based 3D visualization tool to render DEM tiles from multiple angles and compare reconstruction quality in 3D.

**Agentic Job Search Automation Platform**

May 2026 – Present

OpenClaw, n8n, LLMs, AI Agents, Docker, Workflow Automation, SearXNG, Telegram API

- Building an agentic AI system to search jobs, analyze job fit, and generate tailored application materials.
- Designed LLM-based workflows for matching, skill-gap analysis, and application decision support.
- Integrated Google Sheets, Google Drive, and Telegram API to track jobs and send automated application alerts.
- Prototyped the initial workflow in n8n before migrating toward a more flexible OpenClaw-based agent system.

**Persian Text Normalizer for NLP**

Oct 2016 – Feb 2021

Python, NLP, Text Processing

- Developed a Python-based text normalization toolkit to standardize Persian text for downstream NLP pipelines.
- Implemented preprocessing rules for cleaning, formatting, and preparing Persian text data for language-processing tasks.

## Publication

**Amir Mirzai Golpayegani**, Mahmudul Hasan, Faramarz F. Samavati. *Real-Time Multiresolution Management of Spatiotemporal Earth Observation Data Using DGGs*. Remote Sensing, 2025. [doi:10.3390/rs17040570](https://doi.org/10.3390/rs17040570)

## Certificates

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| <b>Supervised Machine Learning: Regression and Classification</b> ,<br>DeepLearning.AI and Stanford University, Coursera | Oct 2024 |
| <b>Applied Machine Learning in Python</b> ,<br>University of Michigan, Coursera                                          | Jul 2020 |

## Languages

**English:** Full Professional Proficiency (TOEFL 115/120)  
**Persian:** Native Proficiency

## Achievement

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| <b>Teaching Assistant Excellence Laureate</b><br>Department of Computer Science, University of Calgary — awarded for outstanding teaching support ( <a href="#">link</a> ) | 2024/2025 |
| <b>Ranked 17th Nationwide in Iran University Entrance Exam (Konkour)</b><br>Top 0.001% among all participants                                                              | May 2016  |